Miraslau Pisaryk

Java / Spring Boot Backend Engineer | BSUIR Student & Project Portfolio

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github.com/CHIKOJgg | linkedin.com/in/miroslav-pisaryk-953490261 | **Open to Intern / Junior / Graduate Remote Work**

SUMMARY

Computer Science student at BSUIR building production-grade backend systems on Java 21/25 and Spring Boot 3/4. Experienced in engineering high-throughput transactional backends (thread-safe in-memory order matching engine, real-time WebSocket STOMP quotes, Redis caching) and architecting distributed event-driven microservices on Spring AI, Keycloak 26, RabbitMQ, and PostgreSQL 17 + pgvector RAG. Proficient with Docker Compose, Testcontainers, and CI/CD pipelines. English C1 Advanced.

SKILLS

Languages: Java (17, 21, 25), SQL, TypeScript/JavaScript (Foundations)

Java Core: Collections, OOP, Generics, Stream API, Concurrency (Virtual Threads / Loom, CompletableFuture, ConcurrentHashMap)

Frameworks: Spring Boot 3/4, Spring AI, Spring Cloud OpenFeign, Spring Data JPA / Hibernate, Spring Security, WebSocket (STOMP), Resilience4j, Bucket4j, MapStruct

IAM & Security: Keycloak 26 (OAuth2 Resource Server, OIDC, PKCE, JWT validation via JWKS, JIT Provisioning, TokenRelay)

Messaging: RabbitMQ 3.13 (Topic Exchanges, JSON serialization, Dead-Letter Queues, Decoupled Domain Events)

Databases: PostgreSQL 17, pgvector (Vector Embeddings, RAG), Redis 7 (Caching, Pub/Sub, Rate Limiting), Flyway Migrations

DevOps & Testing: Docker, Docker Compose, GitHub Actions (CI/CD), Maven, Linux/Bash, JUnit 5, AssertJ, Mockito, Testcontainers

Architecture: Distributed Microservices, Multi-Module Monorepo, Event-Driven Architecture, DDD, Clean Architecture, REST RFC 7807

Spoken Languages: English C1 Advanced (EF SET 70/100, verified); Polish — beginner

FEATURED ENGINEERING PROJECTS

AI Assistant Klawa — Distributed Event-Driven Microservices Architecture

Java 25 (Virtual Threads), Spring Boot 3.4, Spring AI, PostgreSQL 17 + pgvector, Keycloak 26, RabbitMQ 3, OpenFeign, Redis 7, Testcontainers

github.com/CHIKOJgg/klawa | Multi-Module Maven Monorepo (contracts, core, ai-agent)

- **Microservices Monorepo:** Designed and built an Event-Driven Microservices platform structured as a Maven multi-module monorepo (klawa-contracts, klawa-core, klawa-ai-agent).
- **Core & Virtual Threads:** Implemented klawa-core on Java 25 with Virtual Threads (Project Loom) for high-throughput non-blocking execution across task and entity workflows.
- **Keycloak 26 Security:** Integrated Keycloak 26 as an OAuth2 Resource Server; set up stateless JWT signature verification via JWKS, realm role mapping, and JIT user provisioning.
- **Spring AI & pgvector RAG:** Engineered klawa-ai-agent microservice on Spring AI; built Retrieval-Augmented Generation (RAG) using PostgreSQL 17 + pgvector for semantic search over user tasks.
- **Function Calling (@Tool):** Implemented @Tool capabilities allowing the LLM to autonomously trigger CRUD commands in klawa-core via Spring Cloud OpenFeign with TokenRelayInterceptor (JWT propagation).
- **RabbitMQ Broker:** Configured RabbitMQ 3.13 (Topic Exchange klawa.events) for asynchronous domain events (TaskCreatedEvent, AiTaskAnalysisRequestedEvent); decoupled embedding generation.
- **Resilience & Testcontainers:** Integrated Resilience4j Circuit Breakers, Bucket4j token bucket rate limiting, Redis caching, and isolated tests using Testcontainers (PostgreSQL, Redis, RabbitMQ).

Trading Platform — Banking and Exchange Trading Backend

Java 21, Spring Boot 3.3, PostgreSQL, Redis, WebSocket (STOMP), Spring Security, Flyway, Docker Compose

github.com/CHIKOJgg/spring-boot-trading

- **Core Banking Engine:** Built a multi-currency banking and exchange backend for deposit, withdrawal, transfers, statements, and trading accounts with full audit logging.

- **Matching Engine:** Implemented a thread-safe in-memory order matching engine (price-time priority) for limit and market orders using Java Concurrency lock-free primitives.
- **Redis Caching:** Integrated Redis for order book depth caching and message buffering, reducing database read operations by ~40% under peak load.
- **WebSocket Streaming:** Engineered real-time order book depth and user notification streaming over WebSocket (STOMP) for 200+ concurrent clients.
- **Database & Migrations:** Designed a 13-table PostgreSQL schema with double-entry accounting ledgers and Flyway versioning; automated Excel (POI) and PDF (iText) exports.
- **Automated Testing:** Covered business logic and matching engine concurrency stress tests with JUnit 5, AssertJ, and Testcontainers.

Crypto Portfolio Optimizer — Quantitative Analysis & Backtesting Engine

Java 21, JavaFX, ojalgo, CoinGecko API, Maven | github.com/CHIKOJgg/MarkowitzModel-java-crypto-analysis

- **Quantitative Models:** Engineered 12 alpha models (Momentum, EWMA, RSI, MACD, Bollinger) and 9 allocation frameworks (Markowitz Mean-Variance, Risk Parity, HRP, CVaR, Black-Litterman).
- **Backtesting & Stress Testing:** Built walk-forward backtesting engine with Monte Carlo simulation (GBM) and Brinson attribution, calculating Sharpe, Sortino, Calmar, and VaR/CVaR 95% metrics.

EDUCATION & CREDENTIALS

Belarusian State University of Informatics and Radioelectronics (BSUIR)

Bachelor's Degree in Information Systems and Technologies | 2024 – 2028 | Minsk, Belarus

Verified English Certificate: EF SET — Score 70/100, C1 Advanced (Official verification: cert.efset.org/ru/cAsbkU)